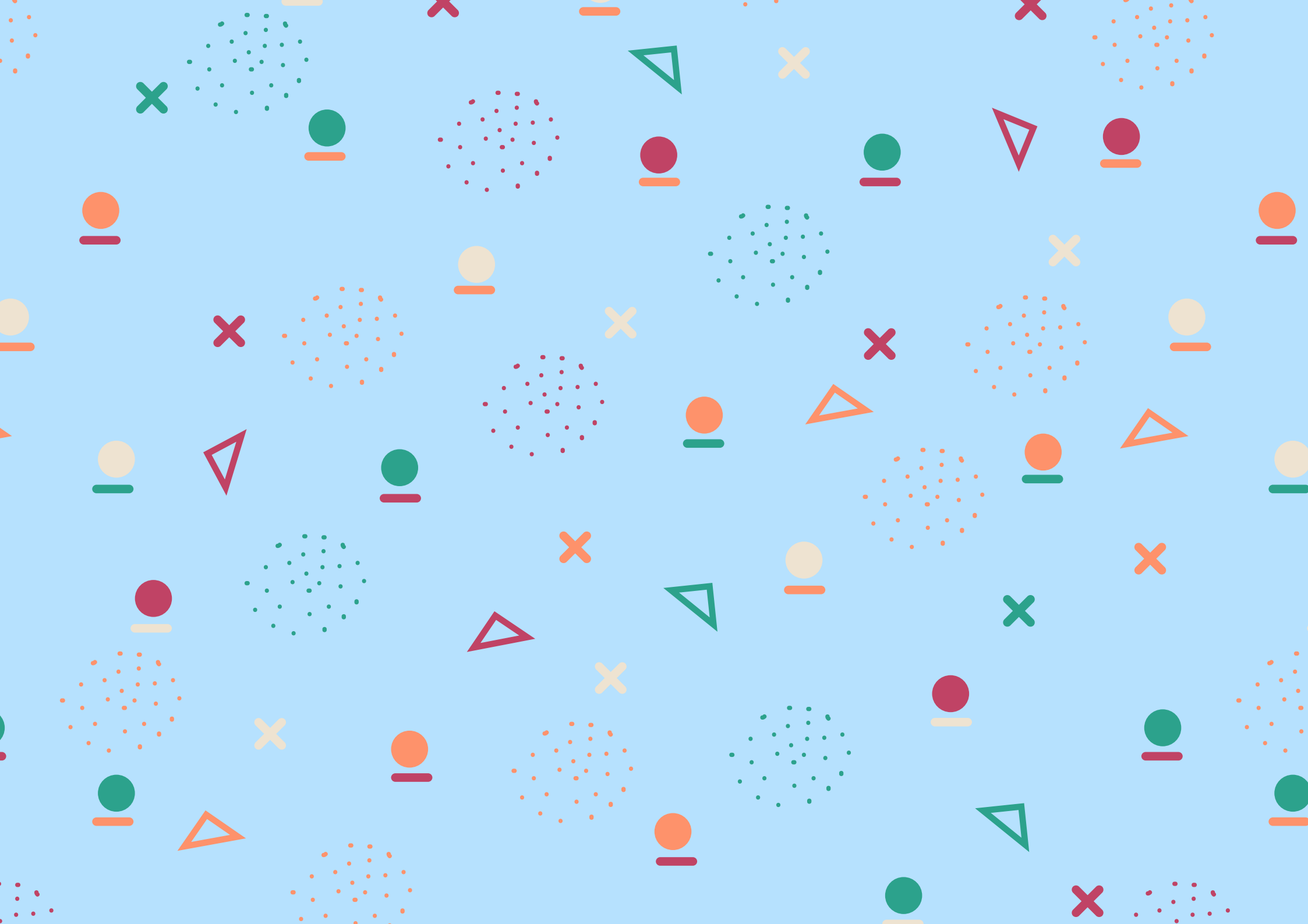


1. If we use nearest neighbor interpolation and bilinear interpolation respectively to get the value of new pixels, please give the values of the points of P1 and P2(both methods).

12				24
	P1			
4			P2	36

Nearest neighbor interpolation: P1=12 ,P2=36; (4%)

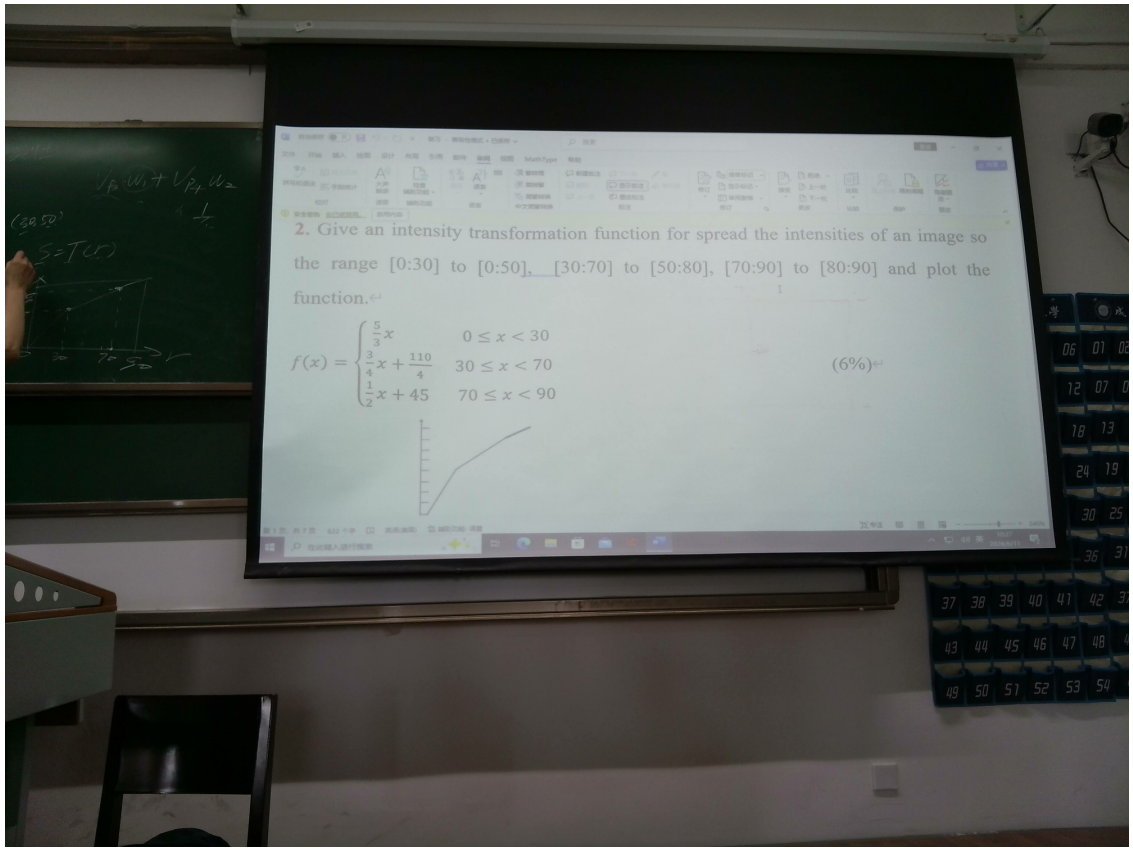
bilinear interpolation:

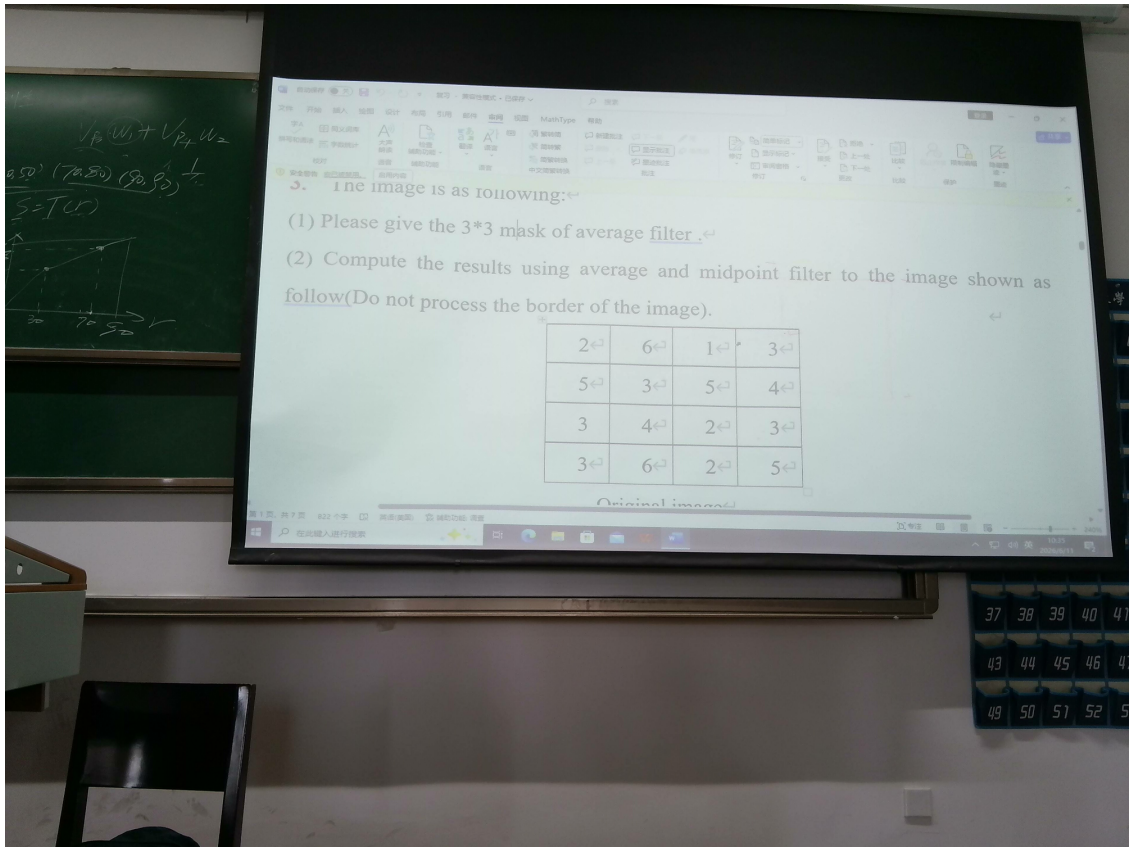
P1=（12*3/4+24*1/4）*3/4+（4*3/4+36*1/4）*1/4=14.25 (3%)

P2=36*3/4+4*1/4=28 (3%)

2. Give an intensity transformation function for spread the intensities of an image so the range [0:30] to [0:50], [30:70] to [50:80], [70:90] to [80:90] and plot the function.

$$\begin{cases} \frac{5}{3}x & 0 \le x < 30 \end{cases}$$


2. Give an intensity transformation function for spread the intensities of an image so the range [0:30] to [0:50], [30:70] to [50:80], [70:90] to [80:90] and plot the function.
$f(x) = \begin{cases} \frac{5}{3}x & 0 \le x < 30 \\ \frac{3}{4}x + \frac{110}{4} & 30 \le x < 70 \\ \frac{1}{2}x + 45 & 70 \le x < 90 \end{cases}$
(6%)

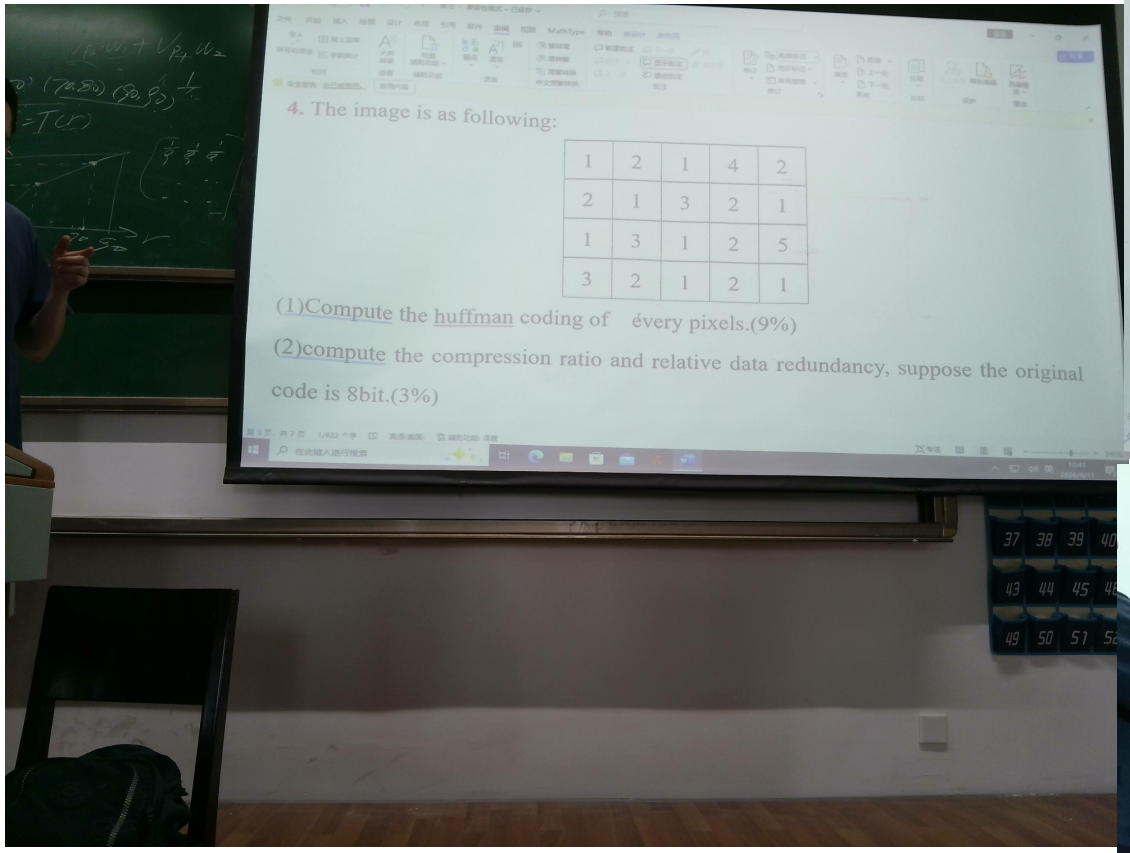


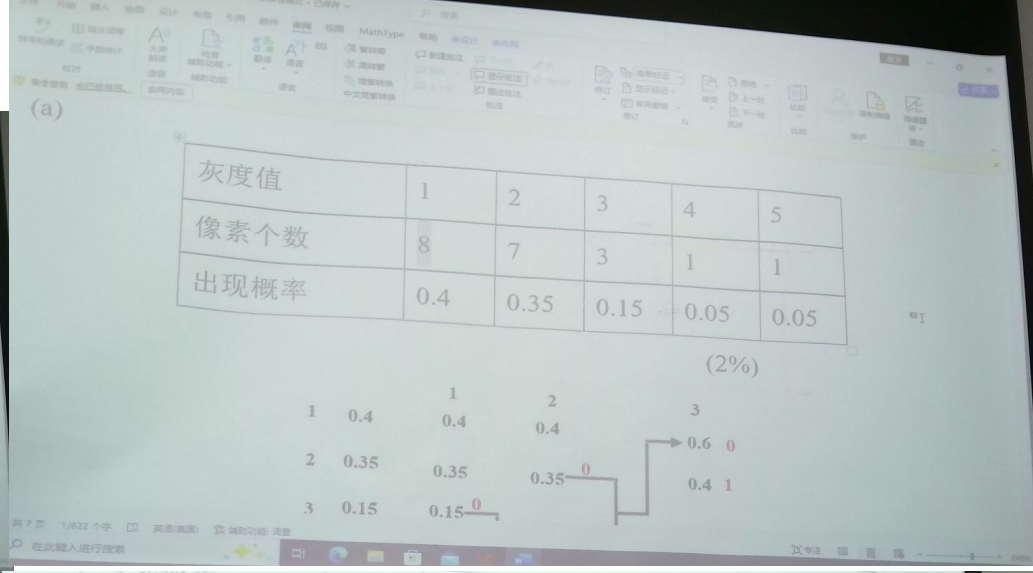
(1) 均值滤波 $\begin{bmatrix} \frac{1}{9} & \frac{1}{9} & \frac{1}{9} \\ \frac{1}{9} & \frac{1}{9} & \frac{1}{9} \\ \frac{1}{9} & \frac{1}{9} & \frac{1}{9} \end{bmatrix}$

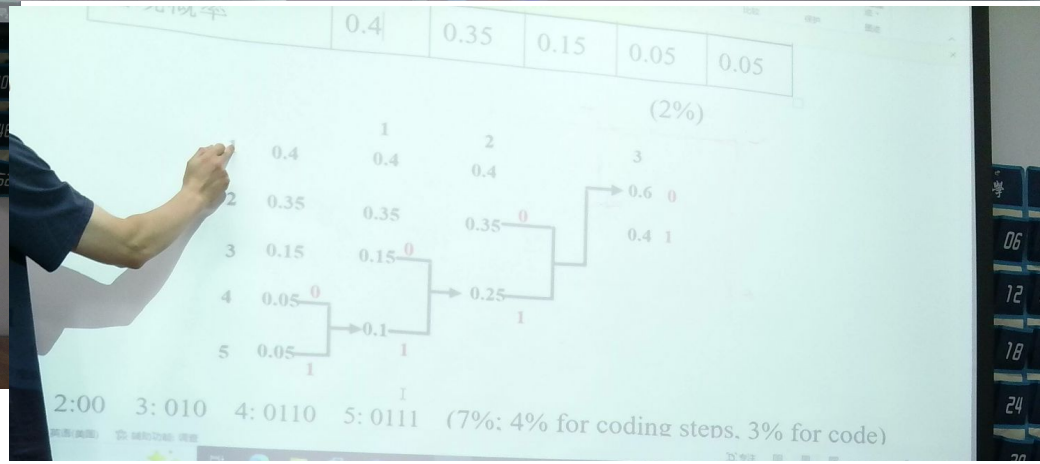
(2) 均值滤波的结果

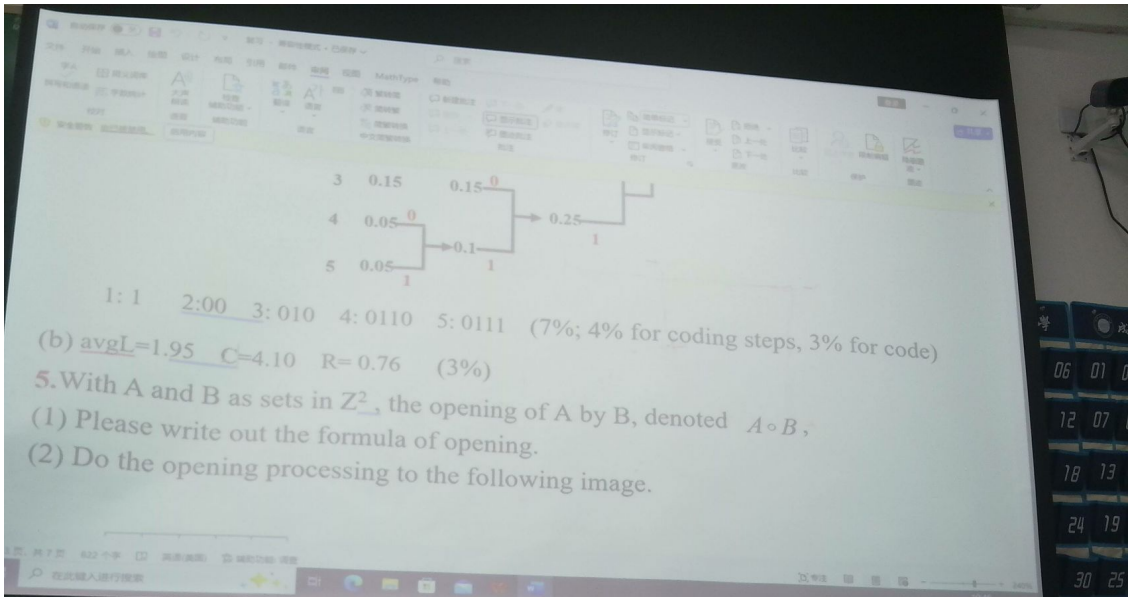
2	6	1	3
5			4
3			3
3	6	2	5

中值滤波的结果


4. The image is as following:
1 2 1 4 2
2 1 3 2 1
1 3 1 2 5
3 2 1 2 1
(1)Compute the huffman coding of every pixels.(9%)
(2)compute the compression ratio and relative data redundancy, suppose the original code is 8bit.(3%)


(a)
灰度值 1 2 3 4 5
像素个数 8 7 3 1 1
出现概率 0.4 0.35 0.15 0.05 0.05
(2%)


0.4 0.35 0.15 0.05 0.05
(2%)
2:00 3: 010 4: 0110 5: 0111 (7%: 4% for coding steps, 3% for code)



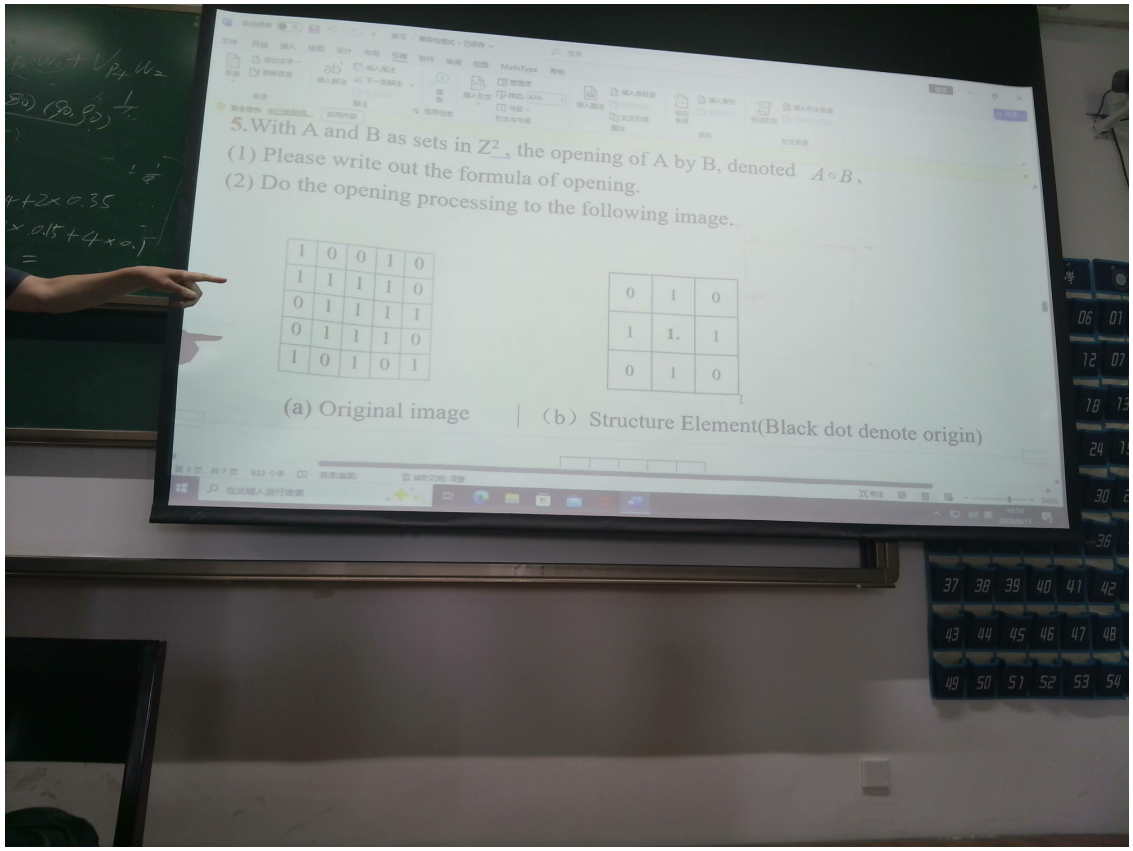
(3) 一张 → 概率

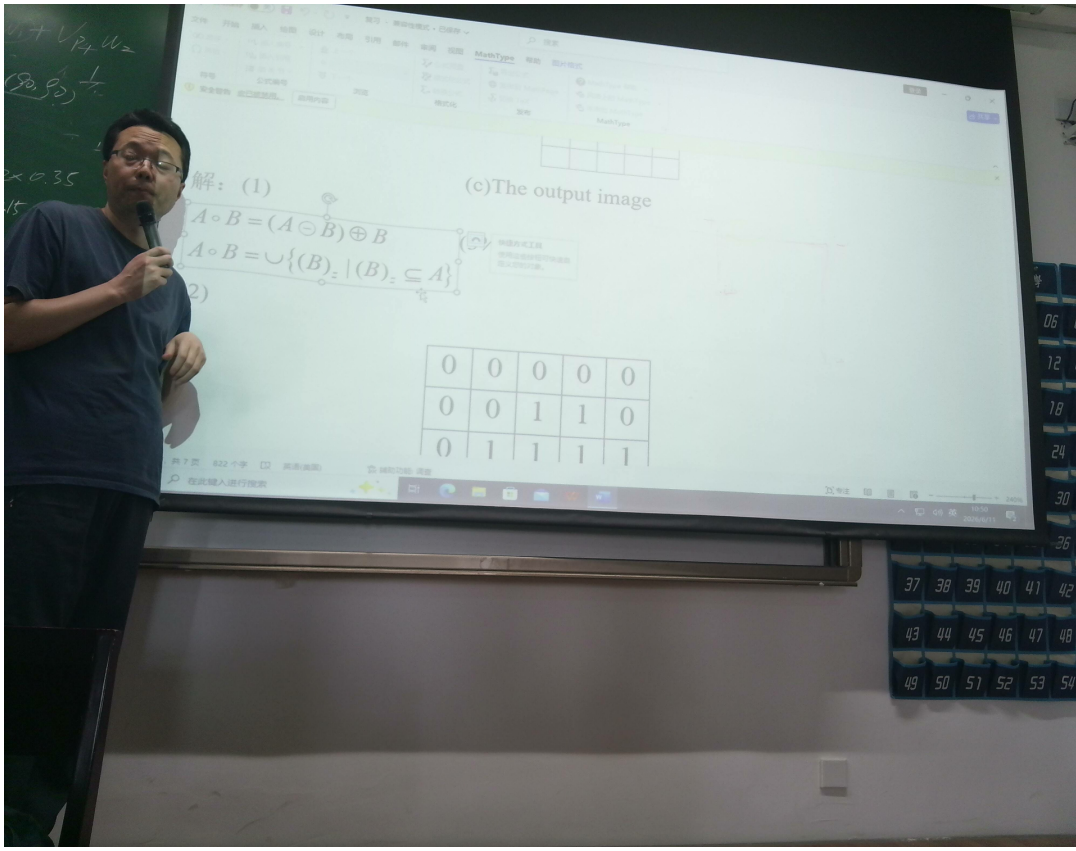
avgL = 1×0.4 + 2×0.35 + 3×0.15 + 4×0.05 + 4×0.05

$C = \frac{8}{1.95}$

压缩率

$R = 1 - \frac{1}{C}$



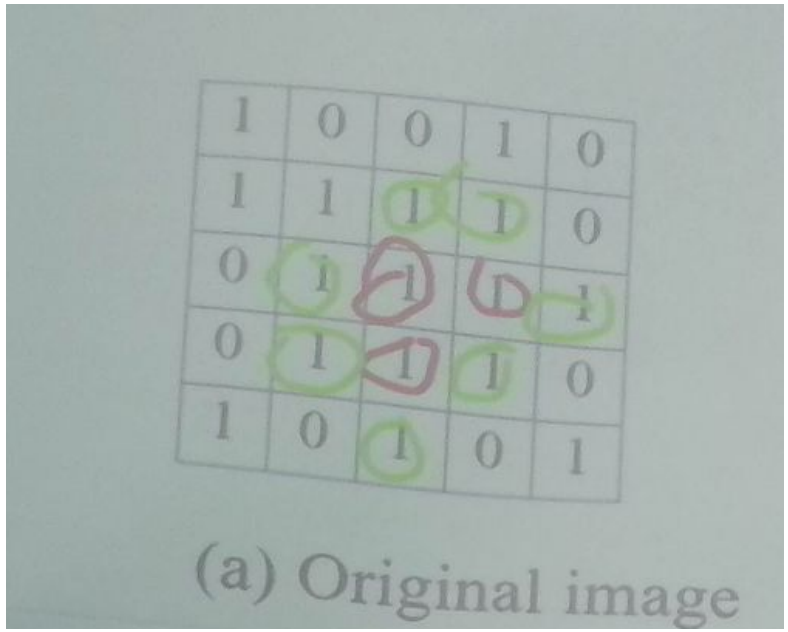


(1) 形态学 腐蚀、膨胀 $A \circ B = (A \ominus B) \oplus B$

(2)

红色是腐蚀

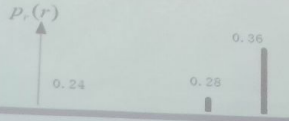
绿色红色 是膨胀

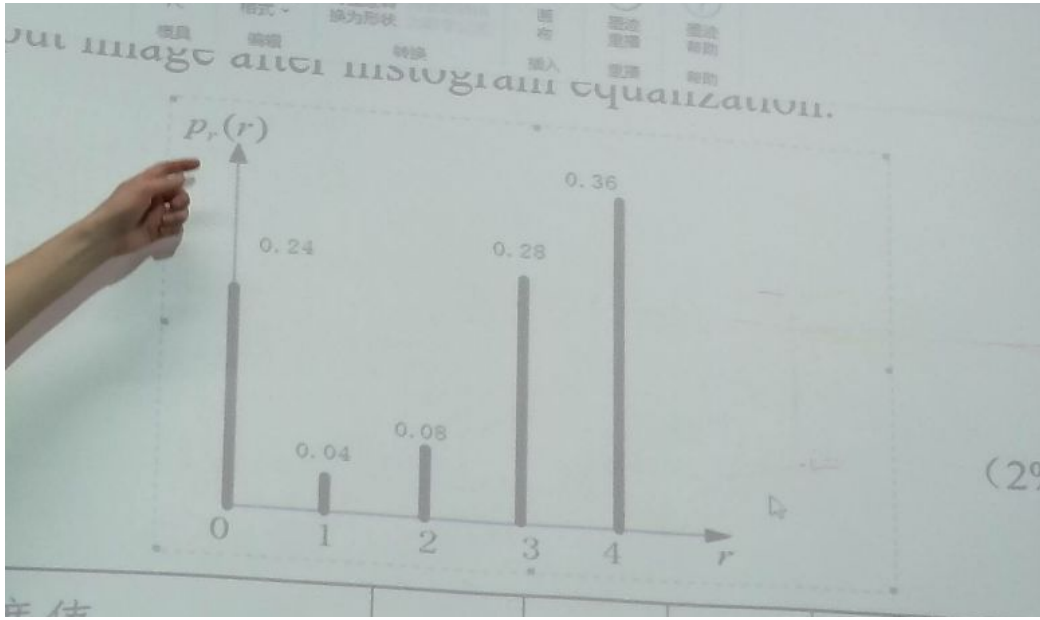


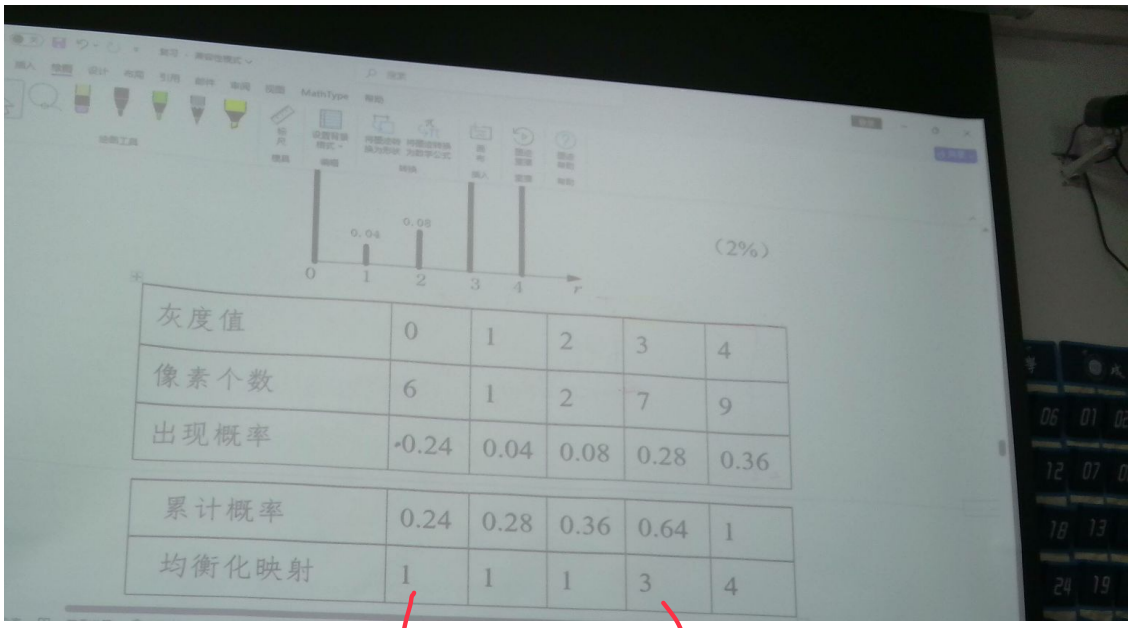
6.An image with 5 gray-levels (the max gray level value is 4) as follow:

2	2	3	3	3
0	3	3	3	4
0	3	0	4	4
0	0	0	4	4
1	4	4	4	4

(1) Plot the histogram of the image.
(2) Process the histogram equalization.
(3) Give the output image after histogram equalization.



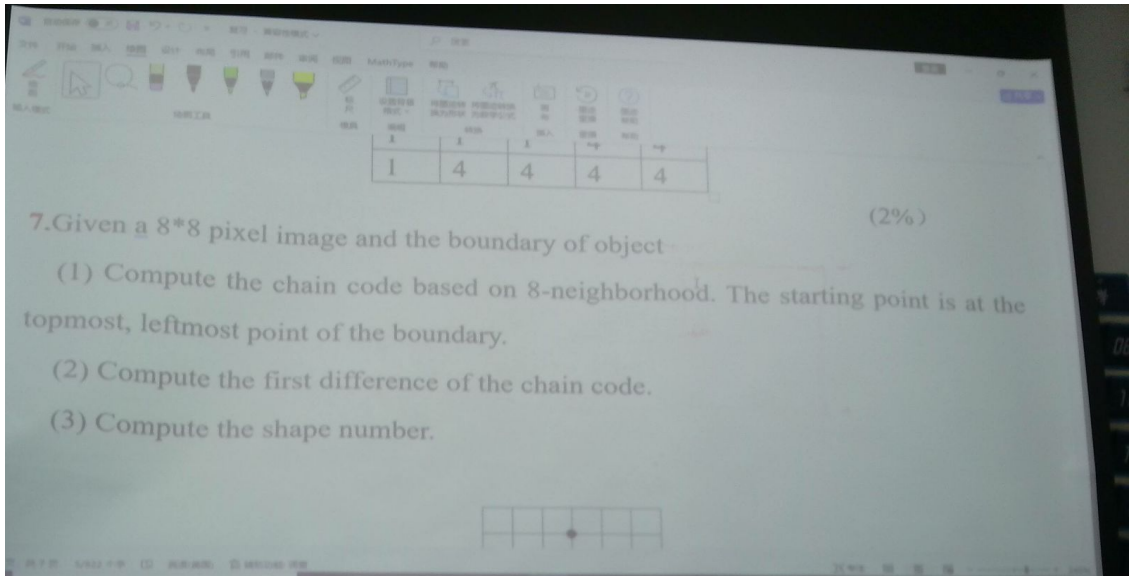




灰度值	0	1	2	3	4
像素个数	6	1	2	7	9
出现概率	0.24	0.04	0.08	0.28	0.36

累计概率	0.24	0.28	0.36	0.64	1
均衡化映射	1	1	1	3	4

0.24×4 取整　0.64×4取整



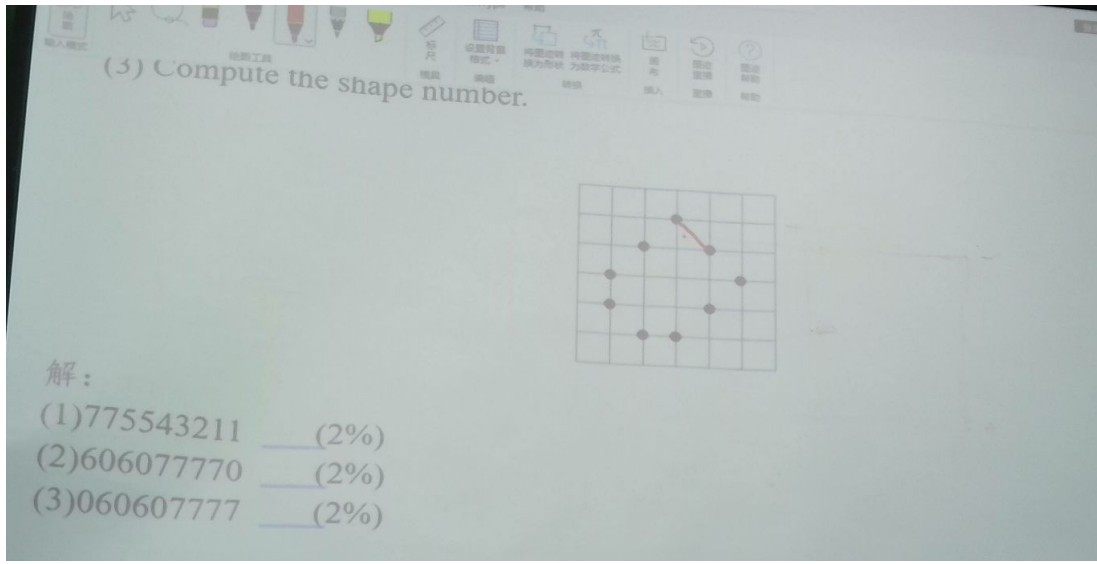
1	4	4	4	4

7.Given a 8*8 pixel image and the boundary of object (2%)

(1) Compute the chain code based on 8-neighborhood. The starting point is at the topmost, leftmost point of the boundary.

(2) Compute the first difference of the chain code.

(3) Compute the shape number.



(3) Compute the shape number.

解：

(1)775543211 (2%)

(2)606077770 (2%)

(3)060607777 (2%)

(1) 链码

(2) 差分 606077770

77

(3) (去掉一位)

8.For any point (x,y) in the image, the response, $g(x,y)$, of the filter is defined as

$$g(x,y)=\frac{1}{6}f(x-1,y)+\frac{1}{6}f(x,y-1)+\frac{1}{3}f(x,y)+\frac{1}{6}f(x,y+1)+\frac{1}{6}f(x+1,y)$$

(1) Give mask of this filter for mask or template operation.

(2)Compute the result using this filter to the image shown as follow(Do not process the border of the image).

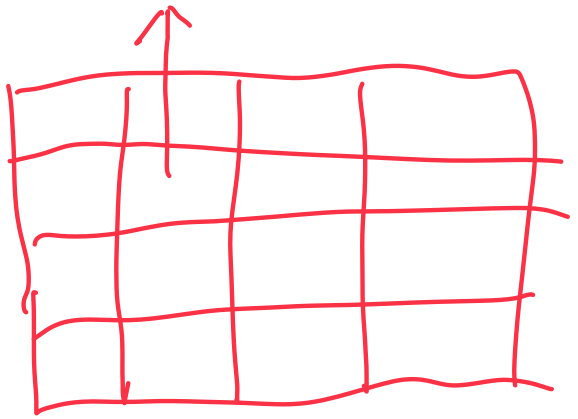
20	7	80	6
11	15	3	2
8	60	220	40

(1)

$$\begin{bmatrix} 0 & \frac{1}{6} & 0 \\ \frac{1}{6} & \frac{1}{3} & \frac{1}{6} \\ 0 & \frac{1}{6} & 0 \end{bmatrix}$$

(2)

$$15\times\frac{1}{3}+(7+11+3+10)\times\frac{1}{6}$$



(2)Compute the result using this filter to the image shown as follow(Do not process the border of the image).

20	7	80	6
11	15	3	2
8	60	220	40
120	8	19	4

20	7	80	6
11			2
8			40
120	8	19	4

9. 基本的全局阈值法（背相应处理步骤）

9. the basic steps of basic global thresholding algorithm

1. Select an initial estimate for the global threshold, T.

2. Segment the image using T in following Eq 1. This will produce two groups of pixels: G_1, consisting of pixels with intensity values >T; and G_2, consisting of pixels with values $\leq T$.

$$g(x,y) = \begin{cases} 1 & if\ f(x,y) > T \\ 0 & if\ f(x,y) \leq T \end{cases} \qquad \text{Eq 1}$$

3. Compute the average (mean) intensity values m_1 and m_2 for the pixels in G_1

pixels with values $\leq T$.

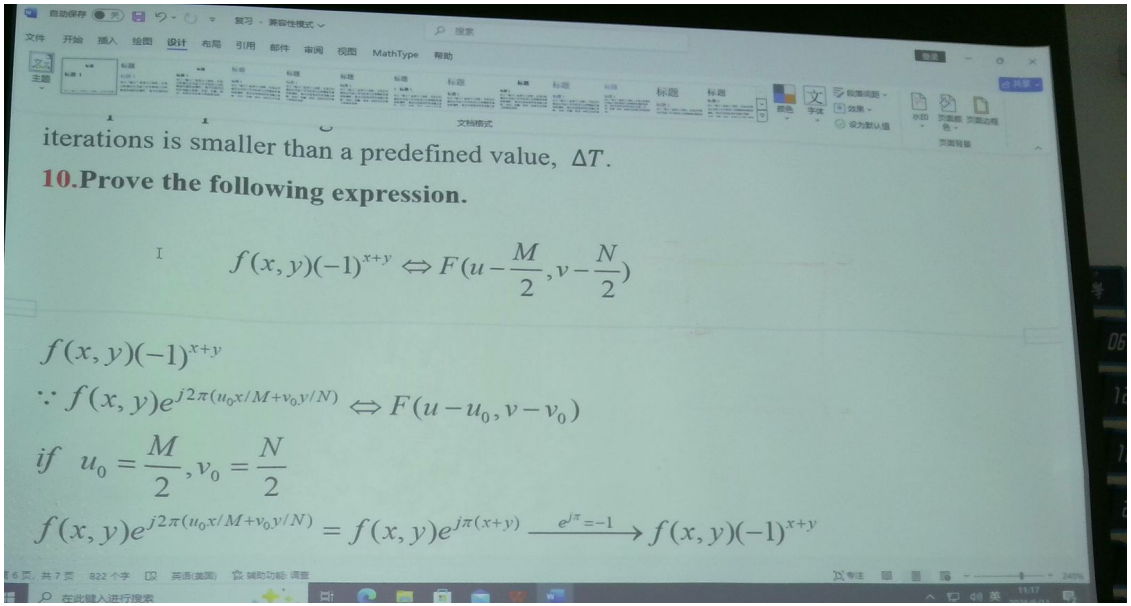
$$g(x,y) = \begin{cases} 1 & if\ f(x,y) > T \\ 0 & if\ f(x,y) \leq T \end{cases} \qquad \text{Eq 1}$$

3. Compute the average (mean) intensity values m_1 and m_2 for the pixels in G_1 and G_2, respectively.

4. Compute a new threshold value midway between m_1 and m_2 :

$$T = \frac{1}{2}(m_1 + m_2)$$

5. Repeat Steps 2 through 4 until the difference between values of T in successive iterations is smaller than a predefined value, ΔT.



iterations is smaller than a predefined value, ΔT.

10.Prove the following expression.

$$f(x,y)(-1)^{x+y} \Leftrightarrow F(u-\frac{M}{2},v-\frac{N}{2})$$

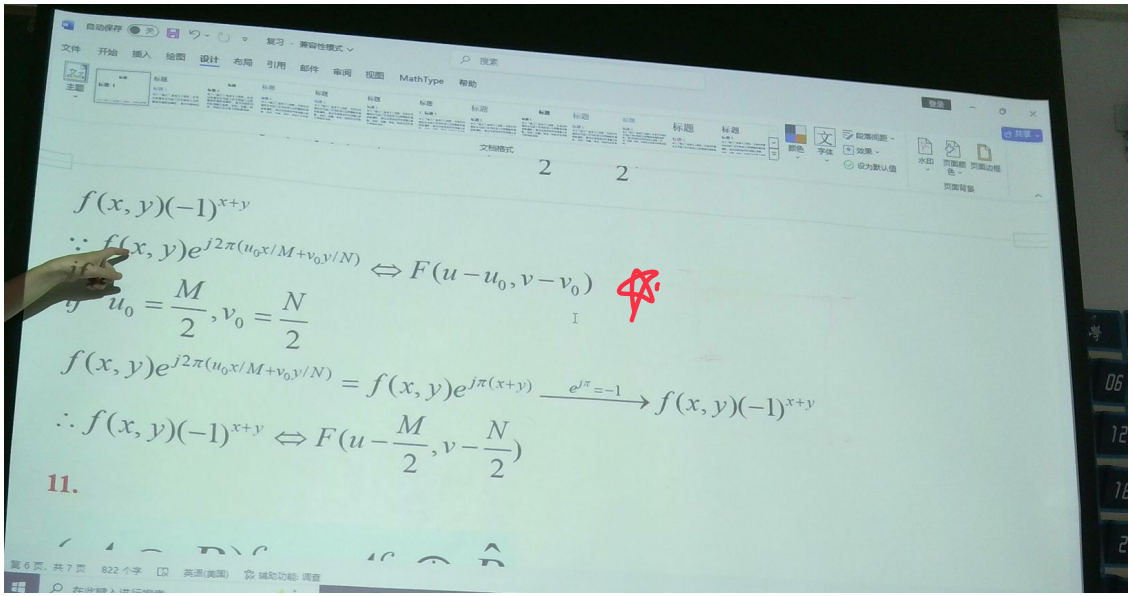
$$f(x,y)(-1)^{x+y}$$

$$\because f(x,y)e^{j2\pi(u_0x/M+v_0y/N)} \Leftrightarrow F(u-u_0,v-v_0)$$

$$if \quad u_0=\frac{M}{2},v_0=\frac{N}{2}$$

$$f(x,y)e^{j2\pi(u_0x/M+v_0y/N)}=f(x,y)e^{j\pi(x+y)} \xrightarrow{e^{j\pi}=-1} f(x,y)(-1)^{x+y}$$

10. 证明 频域



$$f(x,y)(-1)^{x+y}$$

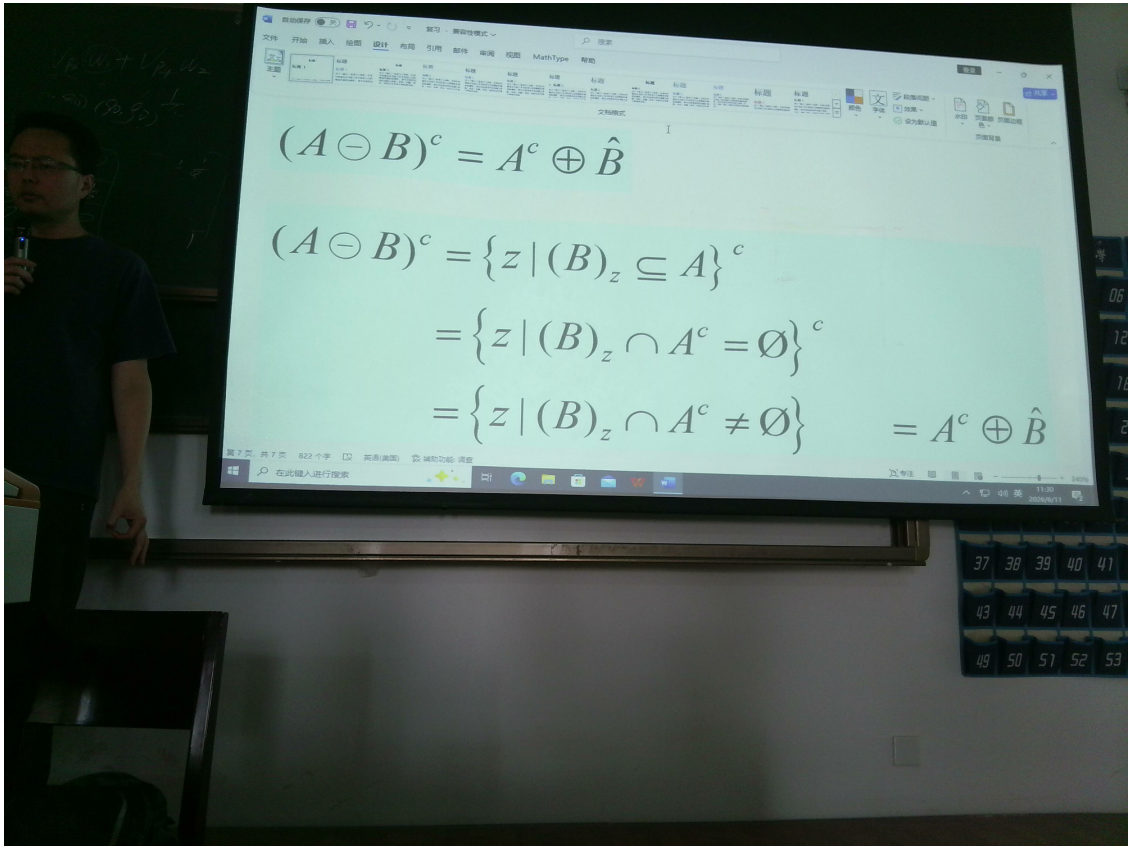
$$\because f(x,y)e^{j2\pi(u_0x/M+v_0y/N)} \Leftrightarrow F(u-u_0,v-v_0)$$

$$if \quad u_0=\frac{M}{2},v_0=\frac{N}{2}$$

$$f(x,y)e^{j2\pi(u_0x/M+v_0y/N)}=f(x,y)e^{j\pi(x+y)} \xrightarrow{e^{j\pi}=-1} f(x,y)(-1)^{x+y}$$

$$\therefore f(x,y)(-1)^{x+y} \Leftrightarrow F(u-\frac{M}{2},v-\frac{N}{2})$$

11.

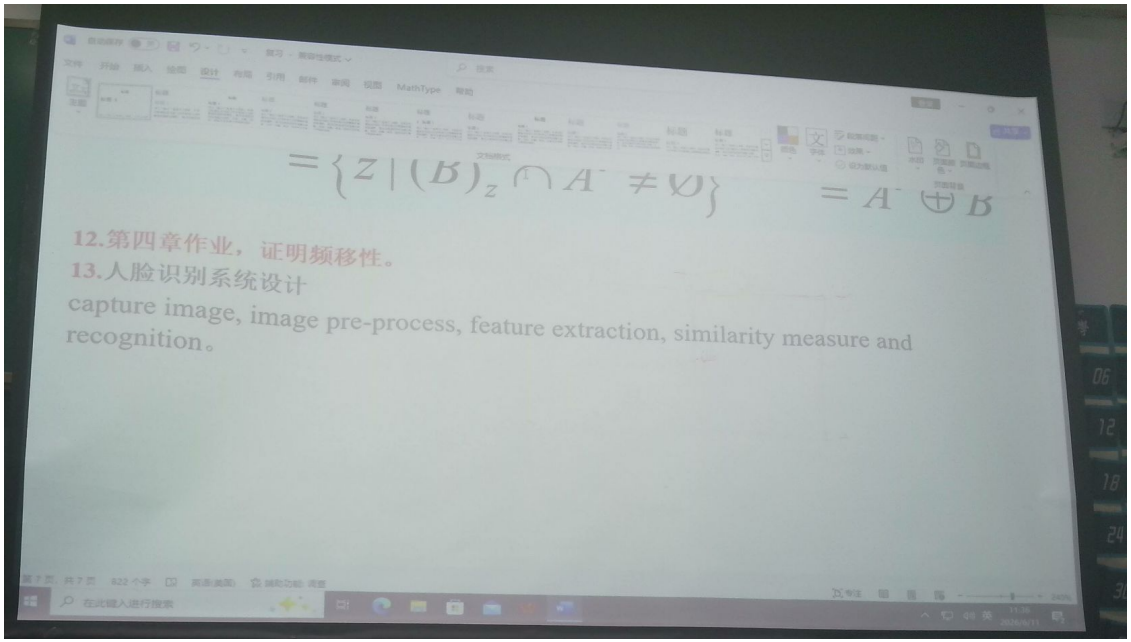


A六边形腐蚀出的补集=A的补集

$(B)_z$ B做平移

$A \oplus B = \{z \mid (\hat{B})_z \cap A \neq \emptyset\}$.

$\hat{B}$ B做映射.



综合题页：

颜色的阈值

眼睛的宽度.